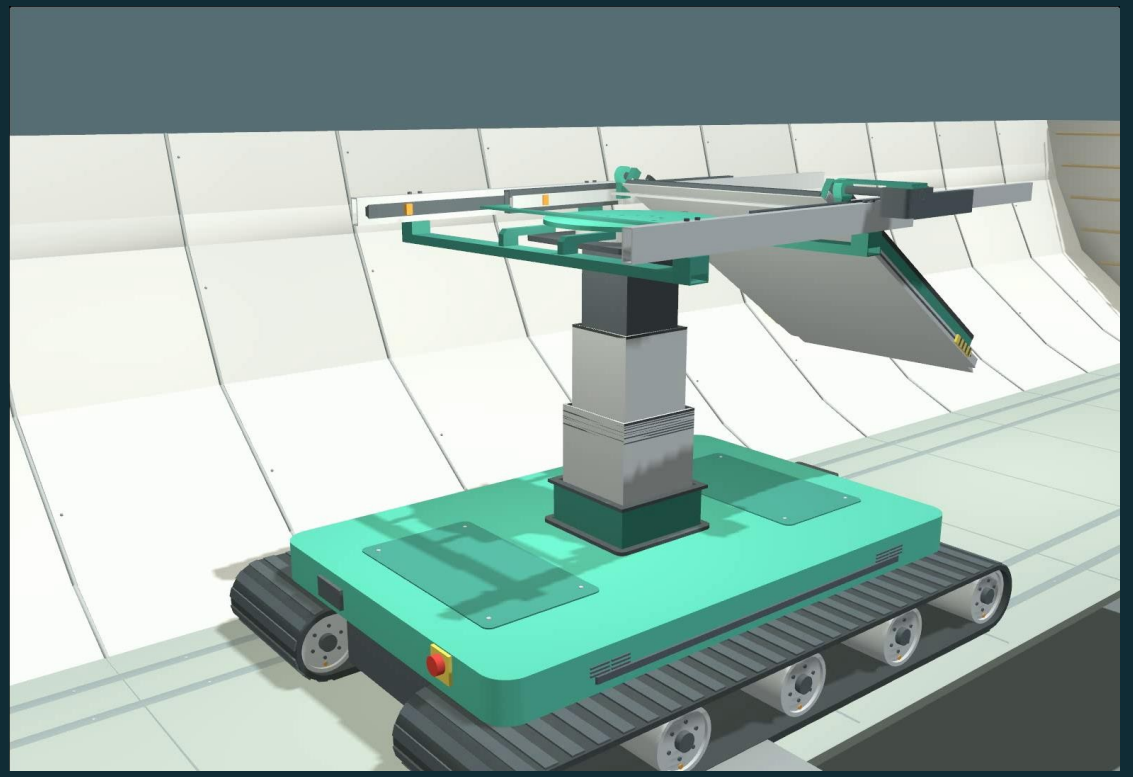
ROAR LAB OPEN HACKATHON 2026

BellyBot

Autonomous baggage stacking and recovery in narrow-body aircraft cargo holds



Design concept and reported simulation results

Sunda Empire

Batam State Polytechnic

The cargo-hold handling problem

Loose baggage requires repeated handling in a confined hold.

< 1.2 m

Working height

Low-profile hardware and controlled overhead clearance.

20–32 kg

Baggage mass

Irregular cases, fragile contents and loose straps.

≥ 12

Bags per minute

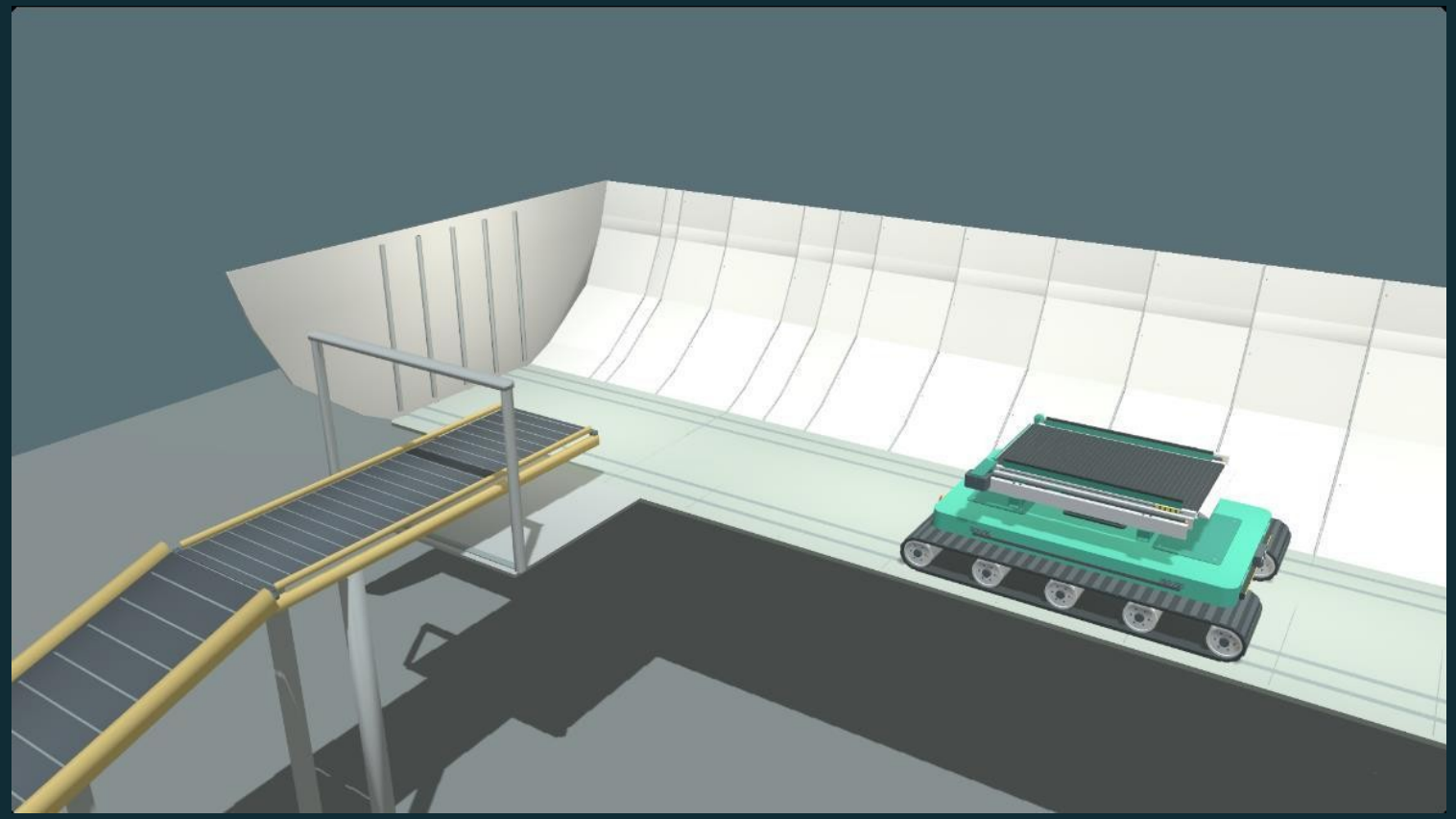
The target covers sustained handling performance.

≤ 600

kg/m² floor loading

Mass must spread through a verified contact footprint.

Door-docked shuttle architecture



20 mm

Nominal belt-to-tray gap

0.498 m

Matched transfer height

A single doorway belt feeds the robot. The design removes stationary conveyors from the hold corridor.

A tracked base with four handling motions

Element	Reported design range	Handling function
Tracked base	1.050 m track length 0.880 m stated width	Shuttle between the doorway and cargo rows
Telescopic mast	0–540 mm lift	Match the tray to the baggage tier
Yaw turntable	±30° stacking ±90° docking	Target side bays and align with the door belt
Reach and pitch deck	0–760 mm reach 0–33° downward pitch	Place baggage and scoop from the floor

32 kg design payload

Bidirectional belt: up to ±0.45 m/s

SYSTEM ARCHITECTURE

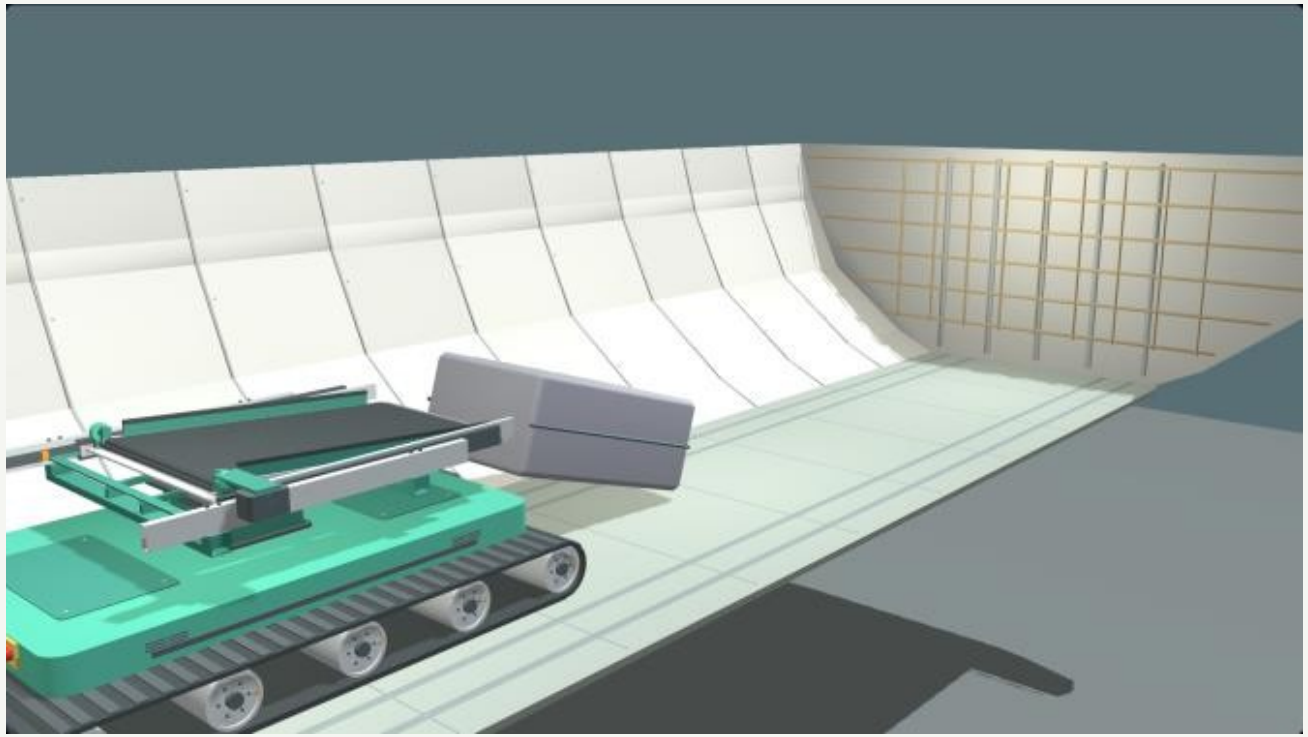
Supervised autonomy and motion guards

Layer	Responsibility
1 Mission controller	Door docking, loading, stacking and recovery sequences
2 Motion guard	Ceiling and sidewall checks, pitch-first retraction, stability screens
3 ROS 2 middleware	Commands, joint feedback, status and simulation clock
4 Gazebo Harmonic	Contact, friction, gravity and joint dynamics

Operator at the doorway

Touchscreen task control, emergency stop and supervised reset

Inbound loading and two-tier stacking



Reported demonstration: six bags, two rows, two tiers

01 Receive and settle

Meter one bag across the doorway belt.

02 Transit to the bay

Pivot into the aisle and follow its centerline.

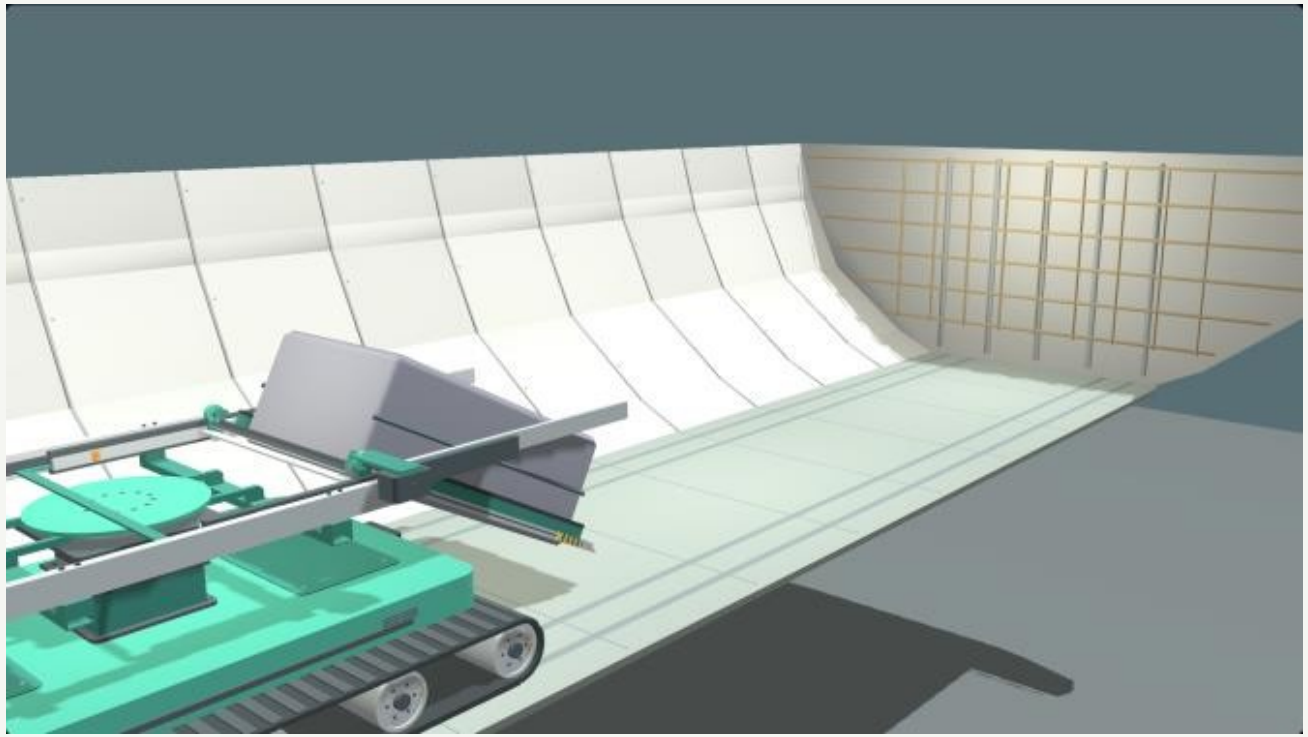
03 Place on either tier

Coordinate lift, yaw, reach and deck pitch.

04 Retract and step back

Level the deck first, then retract and reposition.

Outbound recovery by ground scooping



2.8 s reported acquisition of a 32 kg rigid suitcase

01 Approach the baggage

Move to the designated retrieval datum.

02 Scoop onto the tray

Pitch 29°, extend 680 mm and run the belt.

03 Return to the door

Level and retract the deck before transit.

04 Discharge outward

Reverse the tray belt onto the doorway conveyor.

CARGO-HOLD COMPATIBILITY

Aircraft fit and reported clearances

Constraint	Reported design or simulation evidence
Hold height below 1.2 m	0.468 m stowed height, 1.038 m elevated envelope
Straight aisle envelope	1.22 m aisle, 0.880 m stated robot width
Roof and sidewall clearance	121.8 mm roof, 680 mm sidewall minima reported
Doorway baggage passage	112 mm net margin for a 380 mm thick suitcase
Door interface	20 mm gap and matched 0.498 m tray height

Aircraft-specific geometry and the full swept envelope remain validation items.

Emergency stop and human supervision



3.62 ms

Reported actuator cutoff
20 ms test limit

6.6 mm

Reported stopping drift
25 mm test limit

Stop latches across nodes. Motion commands stay blocked until a supervised reset.

Floor loading depends on the contact model

173 kg robot + 32 kg bag = 205 kg operating mass

Contact assumption	Area	Mean floor loading
Report load-spreading model	0.614 m²	333.6 kg/m² reported
Two stated track shoes	0.378 m²	542.3 kg/m² calculated
Hackathon limit	—	600 kg/m²

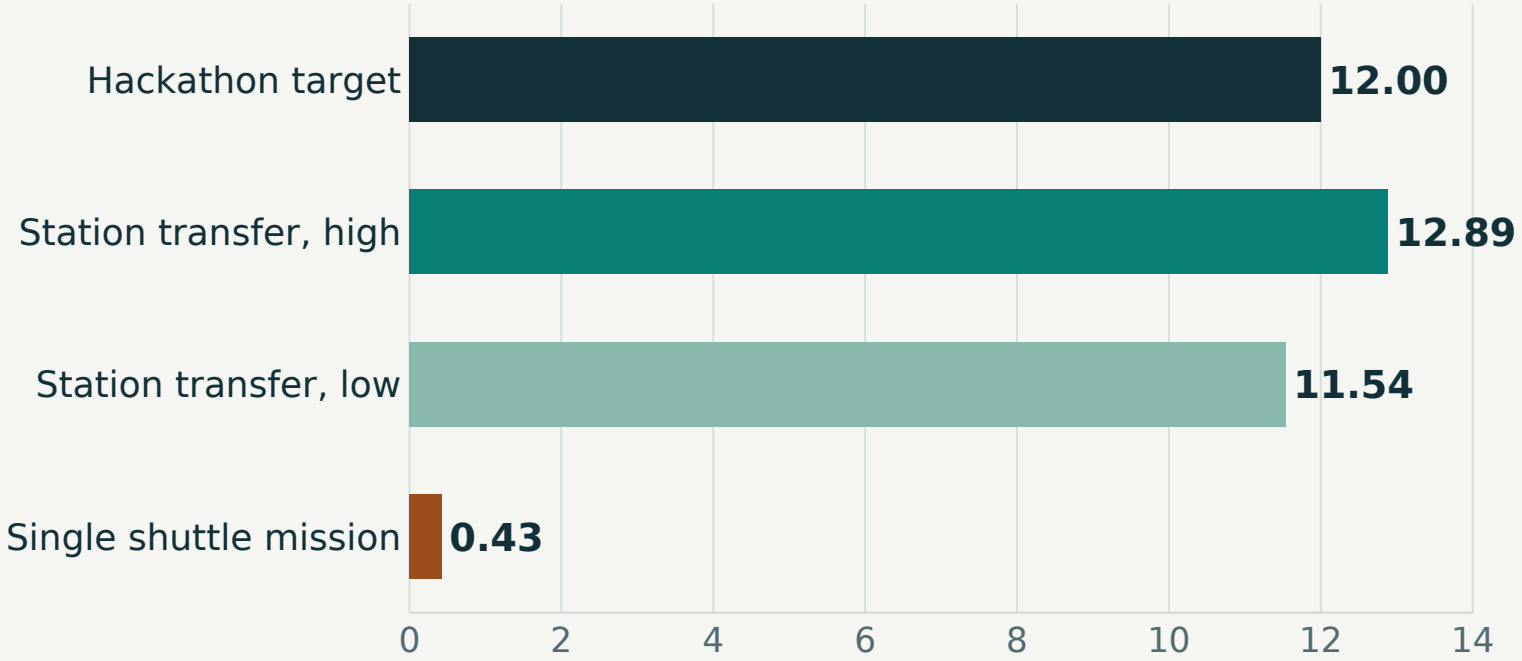
The nominal track estimate leaves about 9.6% below the limit.

Physical tests must establish actual contact, local peaks and dynamic loads.

PERFORMANCE REALITY

Transfer cadence and mission throughput

The complete shuttle mission remains below the ≥12 bags/min target.



140.5 s

Reported complete single-shuttle cycle

0.43

bags/min sustained

Station results measure active transfer only. Their range crosses the target.

Development priorities

01 Physical baggage handling

Test soft bags, straps, off-center loads and fragile contents.

02 Perception and localization

Replace ground-truth poses with sensing and measured pose error.

03 Aircraft envelope and floor tests

Check swept volume, turning clearance and local track loads.

04 Continuous-feed throughput trial

Evaluate an extendable feeder to reduce repeated shuttle travel.

Coverage of the six judging criteria

Judging criterion	Weight	Presentation evidence
Innovation and creativity	20%	Door-docked shuttle and combined placement / scoop
Functional effectiveness	20%	Loading and unloading sequences, slides 6–7
Feasibility and technical soundness	15%	Mechanisms, floor model and throughput, slides 4, 10–11
NB hold compatibility and space use	15%	Open corridor and reported fit, slides 3 and 8
Autonomy and intelligent operation	15%	Supervised control and motion guards, slide 5
Safety and human interaction	15%	Latched stop and operator reset, slide 9

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APPENDIX A

Operating points used in the report

Parameter	Lower tier	Upper tier	Ground scoop
Mast lift	0 mm	210 mm	0 mm
Yaw	±9.58°	±9.58°	0°
Reach	760 mm	760 mm	680 mm
Downward deck pitch	28.0°	20.8°	29.0°
Belt speed	+0.40 m/s	+0.40 m/s	+0.40 m/s

Retraction interlock: level the deck to 0° before retracting the boom.

APPENDIX B

Technical validation before hardware

Open item	Required resolution
Joint travel limits	Reconcile narrative ranges with the actuator-limit table
Roof-clearance datums	Use one aircraft geometry for elevated height and minimum gap
Track and panel load path	Verify actual contact area and peak local loading
Turning and reach envelope	Check the full swept volume with payload and cables
Baggage and perception	Test flexible cases, straps and sensing without ground truth
Sustained throughput	Measure repeated complete cycles against ≥12 bags/min

These are prototype validation tasks. The report supplies simulation results.